

Review

Speaker Diarization: A Review of Objectives and Methods

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Abstract: Recorded audio often contains speech from multiple people in conversation. It is useful to label such signals with speaker turns, noting when each speaker is talking and identifying each speaker. This paper discusses how to process speech signals to do such speaker diarization (SD). We examine the nature of speech signals, to identify the possible acoustical features that could assist this clustering task. Traditional speech analysis techniques are reviewed, as well as measures of spectral similarity and clustering. Speech activity detection requires separating speech from background noise in general audio signals. SD may use stochastic models (hidden Markov and Gaussian mixture) and embeddings such as x-vectors. Modern neural machine learning methods are examined in detail. Suggestions are made for future improvements.

Keywords: index terms—speech analysis; spectral continuity; Gaussian mixture models; neural networks; intonation; mel spectrograms

1. Introduction

Processing of speech signals has had widespread application in recent years: *automatic speech recognition* (ASR—translating spoken audio into text), *speaker verification* (SV—identifying who is talking), language identification (which language is spoken), emotion recognition, and coding (e.g., *linear predictive coding* (LPC) in cellular telephony) are the most common uses. Given the prominence of multi-speaker conversations in podcasts, broadcasts, entertainment, and meetings, another valuable speech application is *Speaker Diarization* (SD), where a computer algorithm automatically estimates “who spoke when?” from a recorded audio signal [1]. As many recent papers have noted, SD still needs much research [2,3], as performance remains far from ideal.

One application for SD is to annotate a transcription derived from ASR, which would normally output an estimated text based on input speech, but one would not know who had said this text. Another motivation is to assist the ASR process, to allow application of specific speaker-dependent models to input speech for each portion labelled as spoken by a given person. SD can provide speaker identities for conversational artificial intelligence and help to retrieve audio/video recordings from desired target speakers. Applications for SD also include generating meeting/interview transcripts, medical notes, automated subtitling and dubbing, and downstream speaker analytics [4].

The SD field may include rich transcription, textual content retrieval, and indexing from speech; these applications utilize aspects beyond the scope of basic SD, e.g., determining syntactic boundaries and disfluencies (speech errors such as restarts and hesitations). Speech also often contains so-called *vocalizations* (e.g., back-channels, filled pauses, singing, infant babbling, laughter, coughs, breaths, lip-smacks, and humming) that are not directly relevant for ASR. This paper will concentrate on acoustic analysis of speech, rather than on textual matters, and we include vocalizations as forms of speech. Incorporating textual



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information into SD would render it dependent on language, and general SD, as discussed here, can be developed in a language-independent fashion.

Unlike most other speech applications, where objectives are usually straightforward, the task of SD can be phrased in different ways, depending on a priori knowledge of who may be participating in a multi-speaker audio signal. In comparison to SD, the objective for speech coding is to produce a reconstructed audio signal that is natural and intelligible with low transmission rate and latency. For ASR, the aim is the text intended by a speaker. For speaker, language, or emotion recognition, the desired output is simply the respective identity. However, for SD, the difficulty of the diarization task is greatly influenced by the turn-taking behaviour of speakers, in particular, how much people's speech is separated by non-speech or overlap in their speech [5].

Given an audio signal that contains speech, SD estimates an ordered set of times, called *time stamps*, which correspond to places in the audio each time that a speaker starts and stops talking. In addition, SD labels each time portion (*speaker turn*) by who is talking. For some applications, e.g., closed meetings, all participants are known to the system via prior training. However, when applying SD to arbitrary audio or broadcasts, the number and identities of speakers are often unknown. In these latter cases, SD still outputs time stamps, but notes speech portions (i.e., between each pair of successive time stamps) with labels of Speaker 1, 2, 3, etc. (including portions with multiple speakers).

Audio often has overlapping speech, where two or more speakers talk simultaneously [6,7]. Indeed, in the so-called *cocktail party* situation [8–10], a microphone may record the speech of many people talking at the same time. Applying SD to such cases may not seem useful, but annotating ordinary conversations is of interest, where people often speak over each other. Usually, such interruptions, which result in portions of audio with multiple speakers, are brief, e.g., less than a few seconds and with only two simultaneous speakers (i.e., the original speaker usually ceases talking once the interruption is noticed). Accurate SD would note all such cases. The SD task is similar to other sorts of auditory grouping, such as auditory scene analysis [11].

This paper reviews relevant ideas for SD, including a description of the nature of speech, methods to process (analyze) speech, how to estimate the times of speaker change, and methods to identify speakers. Major focus is on modern machine learning methods, which now dominate most applications of artificial intelligence, but earlier techniques are also discussed. The paper will not examine the related problem of *continuous speech separation*, which isolates a set of non-overlapped speech signals from an audio stream that contains multiple utterances that are partially overlapped [12,13]. As this paper concentrates on classification (time stamps and speaker identities), it will not address peripheral tasks such as speech enhancement (de-noising and de-reverberation) [14], nor discuss joint SD and ASR [15,16].

The work updates earlier reviews of SD [17–19] that were written prior to the recent wave of neural systems. It is similar to a recent detailed review [20] but is more aimed toward explanations of methods and description of speaker aspects to exploit. We assume one microphone as a source, as multi-microphone situations typically use beam-forming, which is beyond the scope of this paper [21,22].

2. Description of the Acoustics of Speech Signals

In pattern recognition applications, it is not always necessary to analyze input data before doing formal classification; some systems process input data directly, based on models derived from automatic training using suitable examples. Indeed, much of modern machine learning takes this latter approach. However, given the cost and complexity of such methods and large models, it is useful to note characteristics of objects to recognize, in

efforts to simplify classification methods. Thus, we discuss here well-understood features of speech signals. These may help estimate points in time where audio shifts from one speaker to another and help identify these speakers.

Natural speech is generated by a speaker forcing air from the lungs through the passages between glottis and mouth, called a *vocal tract* (VT). There are two major classes of speech sounds: voiced and unvoiced. *Voiced* sounds derive from vibrations of the speaker's vocal cords, which create quasi-periodic puffs of air that excite the VT, which acts as a filter to shape the spectrum of the sound. Waves of air pressure exiting one's VT appear as repeated *pitch periods*, but these deviate slightly from true periodicity, as muscles do not move with exact repetition. *Unvoiced* speech, on the other hand, is random noise generated by passing air through a constricted portion of the VT. The vast preponderance of sounds in the world's languages fit into these two voiced and unvoiced classes (some languages also have clicking and sucking sounds, but these are far less common).

2.1. Aspects of Speech to Help Estimate Time Stamps

In terms of the temporal waveform of speech, the signal consists mostly of voiced sounds, which appear as patterns that roughly repeat with every closure of the vocal cords; this abrupt action generates a wide spectral band of energy that acts as a rough "impulse" to excite the VT filter. As the VT has a mostly tubular structure that is much more narrow (e.g., 1–2 cm diameter) than its length (average of 17 cm for men), the spectrum of the VT output is that of a series of resonances. For a 17 cm VT, these resonances, called *formants*, are spaced roughly at averages of 500, 1500, 2500, and 3500 Hz. These values derive directly from the speed of sound (about 340 m/s), in a quarter-wavelength resonator, which is an approximate model of a VT closed at the glottis and open at the lips [23]. In strong voiced sounds called *sonorants*, which dominate most of speech, the intense lower formants slowly rise and fall with VT shape changes. These resonance movement contours, which last from 60 ms up to a few seconds (mostly 100–300 ms), are distinctive for both ASR and SD. However, few modern classification algorithms exploit such analysis, as attempts to track formants were largely discarded from consideration decades ago (with some exceptions: [24]).

Voiced non-noisy sections of speech, thus, have distinctive, regular patterns for individual speakers that can be exploited when judging continuity in audio signals. Both resonances and the vocal cord vibration rate (the latter called the *fundamental frequency*, F_0) change relatively slowly, as speakers utter a sequence of linguistic units called *phonemes*, which are the constituents of words. Typically, phonemes average about 80 ms in duration, while F_0 in men averages about 100 Hz (F_0 depends on vocal cord size and can be up to 1000 Hz in small infants). Thus, a typical voiced phoneme has about eight pitch periods, which slowly vary as F_0 changes and as the VT moves from phoneme to phoneme (motion called *coarticulation*).

For SD, abrupt audio changes are often relevant. Except when the VT excitation changes between voiced and unvoiced (which occurs, on average, 0–2 times per second), the speech spectrum is often fairly stable, with incremental changes from period to period. The other source of abrupt spectral change in speech occurs for nasal consonants, e.g., when the velum lowers and there is a complete closure in the oral portion of the VT (or the reverse change at the end of such consonants). These abrupt changes in speech within an individual's voice may not occur for long stretches of speech (e.g., in a famous example used in testing speech coders decades ago: "May we all learn a yellow lion roar"). On the other hand, some unusual utterances (e.g., rapidly repeating a syllable such as /ta/) could have several abrupt changes per second.

In conversations with multiple speakers, each person would have a pattern of harmonics (spaced at multiples of F_0 , which changes slowly), modulated by formants that also mostly change slowly, characterized by their VT shape. One could exploit the resulting general continuity of both F_0 and formants (within one speaker's voice) to help track individual voices [25]. A commonly used (older) alternative SD approach is to estimate sets of speech features every 10 ms (at a very common 100 Hz *frame* rate), and then cluster frames with related features into groups of frames that potentially represent a speaker turn. We discuss the advantages and disadvantages of these different approaches below.

2.2. Speech Aspects to Help Distinguish Among Speakers

Discriminating among people by their speech (i.e., SV) could exploit both their physical and behavioural aspects. People have varied VTs (physical) and use them differently (behavioural). Much of SV has preferred to focus on analysis of the physical via use of average VT spectra, for reasons of simplicity as well as the fact that impostors can imitate dynamic VT movements more easily than inherent VT shapes. However, *spoofing* (copying or synthesizing speech to claim false identity) appears to be a much rarer risk for SD than for SV, as SD is almost always applied to reliable data (e.g., there is much more motivation to falsify SV than SD). Thus, SD should be able to apply physical differences in VTs as well as how they are used.

Average resonance frequencies vary indirectly with a speaker's VT length, and average F_0 varies indirectly with one's vocal cord size. However, these only provide two scalar values, which are far less than needed to distinguish among large numbers of speakers. Such features can help to discriminate among speakers, but these provide nowhere near the biometric precision of fingerprints or retinal scans.

Modern SV and SD use average speaker representations based on Mel spectra, which largely ignore speech dynamics, and also obscure useful F_0 effects. Yet, speakers demonstrate significant variability in F_0 and in the timing of VT movements, which directly affect resonance dynamics.

It is possible also that the phase of speech may be speaker-specific, but SV and SD research has yet to demonstrate that. Anecdotal evidence for the speaker-specific phase could be that people can often be distinguished by a single short sung vowel, which shows little dynamic behaviour, and whose spectral amplitude is well modelled with as few as ten LPC parameters. Such few modelling parameters that amplitude spectra provide are insufficient to distinguish among many speakers.

3. Analysis of Speech Signals

Most speech applications try to extract useful information from input audio via some form of analysis or processing. The focus of speech analysis is generally both F_0 and the amplitude spectral envelope of the VT transfer function, e.g., some version of the series of formants, which can be efficiently noted in terms of their center-frequencies and bandwidths. Speech input, in the form of continuous air pressure waves from a VT, is mapped into a digital sequence by analog-to-digital conversion, for use by digital computers. Input speech data are sampled periodically at a suitable rate (at least the *Nyquist rate*, which is twice the highest frequency in the signal), to preserve information over a relevant frequency range for a given application (e.g., for speech by telephone, up to almost 4 kHz) [26].

While various speech applications have a wide range of objectives, very similar methods have been commonly used for most of speech analysis, to transform high-dimensional speech data (e.g., 64 kilobits/s for basic digital telephone speech: 8000 samples/s at 8 bits/sample) to much lower-dimensional classification (e.g., dozens of spectral param-

eters at 100 frames/s). This readily applies to ASR, SV, language identification, emotion recognition, and SD.

Many recent neural-based SD systems (called *end-to-end*, or E2E) avoid all direct (spectral or intonational) speech analysis, on the assumption that any specific processing that is not directly linked to the final task objective may possibly discard pertinent information. Nonetheless, as much of SD is not E2E and uses some form of analysis, we discuss this here. As SD relies both on local cues (e.g., at speaker changes) and long-range comparisons of speaker characteristics, considerations of analysis should examine long- and short-range scopes.

3.1. Simple Spectral Measures

The time waveform of speech may serve directly as input to SD techniques, as a way to simplify SD algorithms. However, the direct utility of speech in time is limited; owing to the high level of encoding in speech, simple measures of overall intensity and periodicity are useful, but the dependence of speech time waveforms on spectral phase variations, which have little direct link to speech message information, has led most speech classification applications, including SD, to use analysis focussed on amplitude spectra.

The simplest spectral method to analyze data is the *discrete Fourier transform* (DFT). It shows energy versus frequency [26], converting N time samples (inside a temporal window that is shifted uniformly to handle the non-stationarity of speech) into N samples (256–512) in frequency. For most classification problems, a disadvantage of the basic DFT is that it has excessive detail. Additional processing (such as data reduction or pattern extraction) is generally needed.

A variant of DFT that is often used for speech classification is *filter bank energies* (FBEs); they smooth DFT amplitudes over restricted ranges of frequency using bandpass filters. The nonlinear Mel scale is often used in the frequency scale [27]. This has equal-bandwidth filters to 1 kHz, and then spaces logarithmically, which matches the reduced resolution that the human auditory system has at increasing frequency (which, in turn, follows the tapering of the basilar membrane in the inner ear). ASR often uses 20–100 FBEs, which reduces data from a 256- or 512-sample DFT, while at the same time retaining enough spectral information.

3.2. Advanced Spectral Measures

A common approach for speech spectral analysis is the *Mel-frequency cepstral coefficients* (MFCCs) [27]. They use similar filtering as FBEs (as well as the Mel scale), but reduce to a smaller set of parameters, typically 13 for telephone speech. This small number is sufficient to discriminate formants to within 100 Hz (which is approximately the smallest amount used by speakers to distinguish among vowels in most languages). For the broad spectral purposes of SD (which need not identify individual sounds), precision of analysis can be lower, thus allowing fewer parameters, if cost reduction is important.

Another common spectral analysis method, *linear predictive coding* (LPC), uses an all-pole model of the speech spectrum [28], to directly focus on the resonances of the VT (spectral poles, or intense resonances, are far more important perceptually than other frequency aspects). Standard LPC is common in cellular speech telephony, but is rarely used for SD, as the other speech measures we discussed suffice for speech continuity decisions and SV.

3.3. Time Windows

Speech is non-stationary, i.e., changes with time, as speakers utter a dynamic sequence of phonemes and words. For analysis, speech is regularly segmented into successive short sections; within each section, one assumes local stationarity [17]. Typical smooth (lowpass)

windows will yield a representation that is an average of local events. For rapid transient speech sounds, such as plosives that have acoustic events as brief as 1 ms, even brief windows often smear out phonetic effects.

Durations of speech vary greatly in informational purpose, e.g., sometimes, spectral changes owing to significant VT movements are rapid, while, other times, speakers articulate long vowels with little VT (and, consequently, spectral) movement. As a suitable and simple compromise, most speech processing calculates measures over brief time windows, updated every 10 ms. Such acoustic features are usually determined over a range of 20–30 ms, and frames are moved every 10 ms. Rates of 100 frames/s are found for many applications of speech; this rate is a compromise to minimize computation while also modelling VT coarticulation well.

3.4. Intonation Features

In speech, patterns of timing and F0 are formed by speakers to assist listeners to make linguistic decisions [25]. Nonetheless, very few speech classification applications (including SD) exploit intonation, as the range and diversity of intonation are difficult to integrate into frame-based classifiers.

For phone or word durations, automatic estimation is difficult, as ASR methods may not be reliable. Many algorithms exist to estimate F0 [29], but intonational patterns require a wide range of speech frames, whereas VT resonances are relevant over short periods.

Nonetheless, intonation has been used for SD in the past [30–32]. The range and average value of F0 in speech are both speaker-specific, and F0 contours are generally slowly changing, which help with both speaker identity and continuity. In general, the features (both for resonances and intonation) chosen for use in SD have been statistical (e.g., means, ranges, and standard deviations), rather than dynamic (e.g., F0 contours over syllables). It may be insufficient for SD to ignore more global, dynamic feature behaviour, which humans use a lot, simply because it is inconvenient to integrate into modern SD tools. Much of what is known about human speech production and perception has been ignored in modern neural systems for speech applications, because this information has been difficult to exploit in the traditional frame-based approaches.

4. Use of Speech Features for Speaker Diarization

The speech features described above (primarily MFCCs and FBEs, often in a form called a Mel spectrogram) are used in most speech classification applications. For SD, they serve both for time stamp decisions, based on spectral continuity, and for labelling speech by speaker between each successive pair of time stamps. We now discuss how to do this for localized speaker diarization.

One can distinguish two cases: (1) an input conversation for SD with a small set of subscribed speakers (known to the classification system via prior training, e.g., a common scenario for meetings), and (2) arbitrary unknown audio with multiple speakers (where the system has no knowledge about speakers). SD systems may assume that the number of speakers is known a priori or is at least bounded.

In the first case, for each segmented speech portion, the system chooses the speaker (from among the subscribed set) that is the closest match to the data. If no match is found (e.g., a spectral distance that the system estimates for each speaker exceeds a pre-set threshold), the system may allow an outcome of “no match”. In both cases, spectral distance can be measured against a general speaker model called a UBM (Section 5.2). We discuss the case where a portion of speech has multiple speakers later.

Speaker labelling errors include *false rejections* (FR) and *false acceptances* (FA). FR (a *miss*) occurs when a classifier rejects incorrectly a true speaker. FA occurs when the system

identifies the wrong speaker. In SV, a frequent evaluation criterion is *equal-error rate*, a low percentage that balances FA and FR. Many classifiers decide relative to a *threshold*, accepting or rejecting when a distance or similarity measure lies above or below such a value; so, FAs decrease and FRs increase as a function of thresholds.

5. Models to Apply for Speaker Diarization

The results of speech analysis can be applied in various ways for SD. The classical methods used stochastic models (Section 5.1) and simple long-term average models (Section 5.2). Recent SV research has led to the embedding approaches of Sections 5.3 and 5.4. The recent surge of neural systems for machine learning are discussed in Section 5.5.

5.1. Hidden Markov Models (HMMs) and Gaussian Mixture Models (GMMs)

From the late 1970s to the early 2010s, the most common methods for ASR were *Hidden Markov models* (HMMs) [33]. Using a *maximum likelihood* approach, HMMs sought $\max P(T|S)$, i.e., the best candidate text T for input speech utterance S , where T is a sequence of words and S a temporal sequence of spectral vectors at the frame rate. For efficiency (as $P(T|S)$ is highly complex), with Bayes' theorem, one instead maximizes $P(S|T)P(T)$, with $P(T)$ being a language model and $P(S|T)$ an acoustic model. An HMM can model a phoneme, a word, or a longer speech unit. It consists of a sequence of states, each representing a spectral pattern of speech (corresponding to a position of the VT), with timing transitions between states for successive frames. Using an off-line model trained with much speech, HMM transitions are assigned probabilities (to model durations), and a probability density function (pdf, e.g., mixture of Gaussians) is estimated for each state's spectrum. While ASR requires frame-based decisions, SV only needs one binary speaker decision for an entire utterance. As a result, SV usually reduces the HMM approach to use only a (single-state) *Gaussian mixture model* (GMM). Gaussians are commonly used as they need only a mean and variance to specify (their relevance for the central limit theorem is not the reason for their choice).

As SD needs an identification for each speech portion, HMMs are sometimes used in a post-processing stage, to estimate the most likely sequence of speaker activities, e.g., utilizing subspace models or clustering via variational Bayes VB-HMM [34]. Other forms of post-processing include *system fusion*, where multiple SD models combine results [5].

5.2. Universal Background Model

SV typically uses a set of background or *cohort* speakers, to enhance the robustness of the recognizer. Such an average *universal background model* (UBM) represents a typical speaker, against which spectral measures are tested [35]. A common UBM may be a full-covariance or diagonal-covariance GMM with thousands of components and hundreds of *eigenvoices* (diagonal covariance is commonly used to simplify computation, although that assumes uncorrelated vector elements). Posterior scaling can be applied to discourage frequent speaker transitions [36]. A UBM represents data from all speakers and conditions and is used to normalize SV scores.

5.3. Embeddings

Labelling speech turns by speaker identity requires only a choice among a set of known subscribed speakers (case 1 in Section 4), or deciding whether the turn matches a speaker that was identified earlier in the data (case 2). In comparing a model of a known (trained) speaker with that of a speaker turn in SD, one may calculate an appropriate distance and compare this with a threshold. Since representation spaces are usually complex for SV (and many other pattern classification), simple distances (e.g., Euclidean) are often sub-par, even after normalization (e.g., of means, variances, or L2 length).

A common method today to classify a set of data of variable duration (e.g., a speaker turn) is to (1) extract a *speaker embedding* (a mapping from data of any duration to a fixed-length vector) with a model [37], and then (2) use a multi-class discriminative training stochastic classifier such as *probabilistic linear discriminant analysis* (PLDA) [38]. Extraction of speaker embeddings is typically d-vectors or x-vectors, which are based on i-vectors (as noted in Section 5.4), and derived from inner layers of a neural network trained on speaker classification. Speaker embeddings assume that only one speaker is present in each segment, and thus, they do not handle overlapped speech.

5.4. Joint Factor Analysis

High-dimensional general recognizers such as *Support Vector Machines* [39] have been adopted to more complex classifiers called *Joint Factor Analysis* (JFA) and *i-vectors* [40]. In representing a speaker, JFA obtains a low-dimensional set of parameters (the i-vector) as a compact embedding. The means of speaker-dependent GMMs form an aggregate *supervector*. The JFA model acts as Gaussian pdfs on speaker- and session-dependent HMM supervectors, where most variance is accommodated with a few hidden variables (speaker and channel factors). A speaker supervector has four components: speaker-independent, speaker-dependent (*eigenvoice*), channel-dependent (*eigenchannel*), and residual. These high-dimensional supervectors (tens of thousands of dimensions), from stacking first-order statistics for each mixture component, are projected into a low-dimensional fixed-length representation (i-vector, of 200–600 dimensions). The i-vector thus models a speaker by removing the UBM mean supervector and projecting into a low-dimensional space (a *total variability* space).

5.5. Neural (Machine Learning) Methods

Data applications in many diverse fields now primarily use *artificial neural networks* (ANNs) as their main model for processing. ANNs are capable of adapting to wide ranges of data [41]. Their nonlinear processing allows them to learn highly complex patterns, but they are very difficult to interpret and generally require models with huge numbers of parameters and calculation, which often exceed the capacity of low-resource devices (e.g., cell phones). ANNs transform data or recognize patterns, based on automatic training from examples. An ANN maps an input sequence (e.g., speech waveform samples) nonlinearly to an output sequence (e.g., a speaker-labelled set of time stamps).

5.5.1. Basics of ANNs

The fundamental element (called a *node*) of an ANN performs a function similar to the action of biological neurons. The simplest ANN is a *multi-layer perceptron* (MLP), consisting of *layers* (sets) of nodes, each sending their outputs to the nodes of the next successive layer, without feedback. For a classification task such as SD, training a basic ANN may be viewed as optimizing the combined locations of hyperplane boundaries of complex regions in a data representation space. The initial ANN layer receives a vector of speech samples from an utterance. Intermediate layers may provide various features for classification; their numbers of nodes vary greatly and are chosen empirically, and the resulting features are often difficult to evaluate. Reducing their numbers can result in so-called *bottleneck* features [42], which are sometimes interpretable.

In natural neurons, dendrites provide potentials to an axon, which yields a binary output: a brief pulse if the weighted sum of its inputs exceeds a *bias* threshold. For an algorithmic ANN node, output is specified by an *activation function* $y = \varphi(w \cdot x + b)$, where w is a N -dimensional weighting vector, b is a scalar bias value, x is a set of N input values, and $\varphi(\cdot)$ is a scalar nonlinear function. The parameters of each model node (i.e., w and b) specify the location of a hyperplane in a data representation space; the node's binary

output corresponds to either side of this hyperplane. To provide accurate solutions to practical data problems, ANNs generally require many millions of parameters to achieve high performance.

The basic *fully connected feedforward* MLP structure is general and costly in size and computation, as all nodes in each layer connect to all nodes in the next layer. Useful information in most data (for speech as well as other signals) is generally spread non-uniformly; thus, a fully connected network is inefficient. ANNs with more than three layers are called deep (DNNs); such detailed architecture assists with handling complex tasks. Most applications use various combinations of the more specific network components described in Sections 5.5.3–5.5.5

5.5.2. ANN Training: Loss Functions and Steepest Gradient Descent

For training network parameters, initial estimates of node weights and biases may be specified at random, or pre-trained on unlabeled data. The ANN learning procedure is iterative, incrementally updating the parameters, based on minimizing a *loss* (cost) function. Direct minimization of network accuracy is not feasible, because ANNs need a differentiable loss function, to allow a product chain of derivatives to show which direction to alter parameters, via *steepest gradient descent*, which guides the model towards local loss minima.

Loss functions approximate a penalty for classification errors. Losses are often chosen empirically and vary greatly among applications [43], but a common one is *cross-entropy*, usually with *softmax* output units or its variants (Angular Softmax) [44]. Softmax uses a normalized exponential function for interpretation as a probability distribution.

All ANNs are trained to minimize a loss function, which is usually chosen to be related to optimizing the task performance. As direct error minimization is often difficult to use (e.g., not differentiable, as needed for gradient descent training), there are many possible losses, which are often chosen based on empirical performance, rather than elements related to the data.

SD typically uses two losses: attractor existence loss and diarization loss. The first loss optimizes the estimation of each speaker's presence (for cases with an unknown number of speakers), and the diarization loss optimizes the active probabilities with a *permutation-invariant* training method [45]. The two losses are summed using a *hyper-parameter* weighting (hyper-parameters are model elements that are often selected empirically a priori). The number of speakers may be estimated with the maximum "eigengap" criterion [46].

5.5.3. Convolutional Neural Networks (CNNs)

For many classification problems, pertinent information is found mostly in small local ranges of data (e.g., for speech, in the center frequencies of high-energy formants). To exploit this, ANNs often have *convolutional* layers of nodes that only process data locally within small regions. If the input has a two-dimensional representation (e.g., for speech, a display of time versus frequency), data values are multiplied by a small square weight matrix (e.g., 3×3), summing (*pooling*) results spatially [47]. One type of CNN applies 1-D convolution (multiplication) in time and is called *time-delay NN* (TDNN); these models have no recurrence (feedback) [48].

So-called *conformers* are composed of four stacked modules, including an initial *feed-forward network* (FFN) module, a multi-head self-attention module (see below), a convolution module, and a second FFN module [49].

5.5.4. Recurrent Neural Networks (RNNs)

To accommodate the uneven distribution of information in speech for both time and frequency, a mechanism that is selective for nodes is useful. CNNs exploit local data

correlations, but recurrence can assist for correlations over longer time ranges. RNNs use feedback and hidden states to store information about past input. Network gates (called input, output, and forget) control the data flow among layers. Typical RNNs are *long short-term memory* (LSTM) and *residual network* (ResNet) [50,51].

5.5.5. Attention

A recent technique called *attention* in networks emphasizes specific portions of data and is flexible to exploit non-uniform distributions of data information. Attention is usually determined as a *correlation* in data sequences using matrix operations that have *queries* (inputs), *keys* (features), and *values* (desired outputs). Attention can be applied over time, frequency, and network layers [52,53].

Neural systems with attention are often termed *Transformers*. These recent *end-to-end* (E2E) methods do not use recurrence and rely only on attention. In such recurrent networks (e.g., LSTM), input is processed sequentially. A Transformer, on the other hand, does not exploit data sequences, but instead uses temporal positional encodings in a separate embedding. Transformers typically require more computation than other ANNs. E2E modelling methods (taking data input directly without analysis, and using a unified, simple architecture) for SD have demonstrated superiority in handling overlapped speech compared to conventional pipelines based on speaker embedding clustering. However, E2E systems usually require much carefully annotated training data, which may require up to 10 min to manually describe each minute of training data.

5.5.6. Neural Variants of Speaker Embeddings

An approach called the *d-vector* (for “deep vector”) has an ANN process for each data frame and its context is to achieve frame-level embedding. The output activations from an ANN’s final layer trained by speech from a given speaker are accumulated to form a compact speaker representation [54]. A low-dimensional speaker embedding model may be the average of activations from that last ANN layer (viewed as a bottleneck layer). These embedding d-vectors represent utterances in a space of fixed dimensions, thus allowing simple classification.

X-vector is another speaker embedding variant that uses a sliding window of frames as input, with a TDNN [55]. It has frame-level and segment-level sub-networks, merged with a *statistics pooling* layer, to find means and variances of frame-level embeddings. This layer projects variable-length input into a fixed-length model by accumulating statistics of hidden node activations. Pooling over frame-level representations concatenates means and standard deviations over each set of speaker data. The x-vector usually has PLDA to calculate scores, while the d-vector uses cosine similarity [56]. X-vector clustering may also be used with variational Bayesian re-segmentation [17,37]. This refines the positioning of the time stamps that were roughly estimated during clustering.

6. Speech Activity Detection (SAD)

A major task in SD is to determine which portions of an input audio signal contain speech, versus having silence or non-speech sounds. This is called speech activity detection (SAD) or voice activity detection (VAD) [57]. Classical approaches used a combination of energy estimation and spectral analysis to carry out SAD. When audio contains speech with only a small amount of background or channel noise, SAD could be as simple as applying an energy threshold, i.e., judge any strong signal as speech. In many practical circumstances, however, a microphone could receive many different sound sources. Since speech has several acoustic features that distinguish it from other sounds (Section 2), energy

should be augmented by applying other useful data features, to accomplish more accurate SAD.

SD research often reports results for two tasks: (1) with “oracle SAD,” which indicates that the classification system knows a priori the actual time stamps (*ground truth*), or (2) the system uses a speech activity detector. A classical approach to speaker-turn segmentation is to use hypothesis testing on acoustic segments with sliding (possibly overlapping) successive windows. Such speaker-change detection often requires a threshold that is empirically tuned for major changes in audio, e.g., movement of the sound sources or the microphone.

As noted in Section 2, speech can be voiced or unvoiced. Unvoiced sounds are weaker and more easily confused with other audio, especially background noise; as such, speech derives from noise generated at a VT constriction but shaped by the upper VT (forward of the constriction). This shortness of the oral cavity of the VT for obstruent sounds generally renders the speech noise as high-pass in nature, as opposed to most environmental noise.

In the majority of speech that is voiced, the resonances of the VT impose a formant structure to the spectrum, being roughly spaced (e.g., 1 kHz apart, on average, for the typical 17 cm long VT of men). In addition, the quasi-periodic glottal excitation of the VT imposes a rough line spectrum, spaced at F0 Hz. Both of these acoustic features make voiced speech unlike many other non-speech sounds. Most SD exploits these features very indirectly, via Mel spectrograms, rather than any direct feature extraction. Most modern SD, based on ANNs, does not explicitly utilize the speech production features noted in this paper, but allows the network models to learn distinctions via automatic training. The general assumption is that it is best to leave processing to the automatic training, but training with simple gradient descent on generic losses imposes a heavy burden on such methods.

7. Clustering Frames by Speaker Turns

In sections of speech, as perhaps estimated by SAD, SD must determine (1) whether one or multiple speakers are talking and (2) at what times speakers start and cease talking. One common approach is to seek times of pertinent change in speech features and/or group (cluster) related speech frames into coherent speaker segments [43,58–61]. Section 2 described features of speech that may be exploited to seek speech continuity. However, given the complexity of speech, most approaches to SD have used more general distance measures. Other earlier methods also used HMM-ASR to assist [58]. ASR can assist in modern neural SD as well [62], as ASR can provide textual clues to speaker turns.

To measure the distance between speech segments, there are two common methods for SD: *generalized likelihood ratio* (GLR) [63] and *Bayesian information criterion* (BIC) [64]. BIC searches for change points within a time window, using a penalized likelihood ratio test of whether data are better modelled by a single or different distributions. GLR is a likelihood-based metric: the ratio between two hypotheses (whether successive segments come from the same speaker or not). Also common is the *Kullback–Leibler* divergence, which estimates the distance between two random distributions [65].

Other popular techniques include *spectral clustering* (SC) [51], *k-means clustering* [66–68], *mean shift clustering* [56], and *agglomerative hierarchical clustering* (AHC) [69] to aggregate the regions of each speaker into separate clusters. These tend to use a bottom-up approach that trains multiple clusters or models and successively merges them, reducing the number of clusters until one remains per speaker. SC mostly uses a cosine distance, rather than the Euclidean.

However, many clustering methods can only estimate one speaker per segment and thus cannot handle speech of overlapping speakers. Also, standard k-means clustering

uses squared error as a criterion, which generally does not handle non-Gaussian data well nor data containing imbalanced classes, which are common for speech. Indeed, many SD change-point detections are based on metric-based approaches that assume the distribution of speech features follow a Gaussian distribution, which is often not the case.

SC is a common unsupervised clustering technique based on graph theory [70]. It requires a pairwise similarity score between all pairs of speaker embeddings, resulting in a similarity graph. SC is sensitive to noise in the affinity matrix.

AHC constructs a hierarchy based on distances between features and forms a group. AHC is a common unsupervised clustering technique, which uses pairwise distances between all speaker embeddings, yielding a distance matrix. A cluster hierarchy is generated (a tree diagram called a dendrogram). Each input sample starts in its own cluster, and pairs of clusters are then merged.

Variations on clustering include adding constraints [71,72] and affinity propagation [73], as well as X-means and online re-clustering [66,74]. There is also Viterbi re-alignment, which re-segments an audio stream based on a current clustering hypothesis, with re-training on an alternative segmentation [75,76]. Most clustering in SD is unsupervised [77]. Some SD studies have used mean-shift clustering, which assigns data points to clusters iteratively via modes in a non-parametric distribution [78].

In *deep embedded clustering* (DEC) [79], input features (speaker embeddings) are transformed to be more separable. For differentiability (needed to train ANNs), each embedding has a probability for available speaker clusters, which are iteratively refined using bottleneck features estimated by an autoencoder [80]. Some SD approaches combine segmentation and clustering into one trainable model [77].

8. Summary of Speaker Diarization Methods

We distinguish two different approaches to SD: classical modular and neural.

8.1. Modular SD

Classical SD uses a sequence of steps: (1) a speaker embedding extractor for each segment, (2) a clustering algorithm that merges segments based on the speaker embedding similarity and assigns a speaker label to each segment, and optionally, (3) some compensation algorithm such as variational Bayesian refinement can calibrate the clustering. In such modular SD, several components must be optimized using separate criteria, and overlapped speech is difficult to handle, as unsupervised clustering often assumes that each segment is assigned to only one speaker.

8.2. Recent ANNs for SD

E2E ANN-based SD (EEND) requires labelled data for all the joint speech activities of every speaker at each time frame in a recording [81–83]. It is usually trained on simulated data and then adapted to the target condition with labelled data. Current state-of-the-art systems include encoder–decoder attractor architecture EDA-EEND [84], x-vector clustering [55], the self-attentive EEND (SA-EEND) model [52], and Region Proposal Network-based SD [48]. Some of these deal with the problem of determining the total number of speakers in a conversation. A limitation of EEND is its fixed number of output heads, i.e., it is difficult to handle varying numbers of speakers. EDA-EEND uses an LSTM encoder–decoder to predict an *attractor* for each speaker (to group frames). Re-segmentation is an optional step that further refines the diarization prediction [34]. EEND has difficulty with verbal meetings with more than three speakers and long-form audio (e.g., recordings exceeding 10 min). Basic EEND operates in batch processing mode, leading to excessive computer memory with long recordings.

SD DNNs often have multiple recurrent layers and develop them with the *permutation invariant training* technique [45]. However, they only show good performance in constrained settings and do not generalize well to real-world conditions.

9. Ways to Evaluate Speaker Diarization

Measuring success for SD is more complicated than for other speech applications. For example, speech coders and synthesizers seek high intelligibility and naturalness. ASR and SV seek minimal word and speaker identity errors, respectively (all, of course, wish to minimize cost and latency). However, SD involves both timing estimation and speaker recognition; it is not immediately clear which is more important. A common measure of SD success is the *Diarization Error Rate* (DER). DER treats three error types with equal weight: *false alarm* (FA: section of audio labelled incorrectly as speech), *miss* (duration of speech not so labelled), and *confusion* (portion of speech with mislabelled speakers). Thus, $DER = (FAs + Misses + Confusions) / \text{Total Duration of audio}$.

DER does not explicitly judge whether time stamps are appropriate, i.e., penalize errors in locating times where a speaker stops/starts; it examines the average of labelling accuracy over all frames. Indeed, some SD challenges allowed error deviations in time stamps via a *forgiveness collar*, which meant ignoring small timing errors (e.g., under 250 ms). This “score collar” around every boundary of reference segments was intended to mitigate effects of inconsistency in reference transcriptions.

Other SD measures include WDER, which is a version of the DER measure based on words, rather than audio durations. The *Jaccard error rate* (JER), as in the DIHARD II set, is based on the Jaccard index, a similarity measure used to evaluate the output of image segmentation systems. $JER = (FAs + Misses) / \text{Total}$; so, it is DER, not counting confusions. DER values typically exceed 10% on most standard datasets (see next section) and can range much higher on some.

10. Datasets for Speaker Diarization

A disadvantage of SD approaches based on models is their reliance on external data for the training, which makes them less robust in acoustic conditions not seen in training. So, data are important. To evaluate SD performance, one typically uses annotated datasets, i.e., sets of audio containing multiple people talking, where time stamps are labelled manually. Training can be carried out on any audio, but supervised training would use labelled datasets, which could be from natural sources (e.g., YouTube, public broadcasts). Training sometimes uses “synthetic” datasets, where individual speakers’ speech (i.e., from monologues, not conversations) are superimposed. Such simulated mixtures often have added noise and reverberation with simulated room impulse responses. SD performance varies with how well the simulated data used in training matches real (target domain) conversations.

Early SD datasets were the ICSI meeting corpus [85] and the NIST Rich Transcription sets from meeting conference audio [86,87]. An earlier dataset still widely used for challenging ASR is the CALLHOME dataset, which has 500 sessions of multilingual telephone speech. Each session has two to six speakers, but most conversations (2–5 min) each have two dominant speakers.

Other datasets are DiHARD [88], Augmented Multiparty Interaction (AMI) Meeting corpus [89], the CHIME series [90–92], Fearless Steps [93], Iberspeech RTV [94], LibriCSS [12], M2Met [95], and the French REPERE [96]. DISPLACE 2023 has conversations in different languages, high levels of noise, reverberations, and overlapping speech.

Freely available datasets include AISHELL-4 [97], Albayzin/RTVE [98], AliMeeting [95], VoxConverse [98], Ego4D [99], and This American Life [100].

The AMI corpus includes 100 h of meeting conversations. DIHARD I and II development sets have 19 and 23 h of data. The VoxConverse dataset has approximately 20 h of conversations from YouTube. The VoxCeleb datasets, widely used in SV, have speech segments of more than ten seconds (on average). An open-source toolkit for SD is written in Python [101].

11. Discussion

This paper has reviewed the technical problems and methods associated with speaker diarization. It first discussed the acoustical nature of speech signals, from the point of view of which aspects or features may be employed to distinguish speakers. Notably, most speech is voiced (quasi-periodic) with resonances and F0 that mostly change slowly. Approximately eight times per second, speech transitions between phonemes, which results in spectral changes that are sometimes abrupt and sometimes smooth. Otherwise, i.e., during approximately 85–90% of speech frames, there is significant spectral continuity in the acoustic signal of a speaker.

Individual phonemes have patterns of two types: (voiced) sonorants or noisy consonants. The former have slowly changing resonances (on average, one per kHz in typical 17 cm vocal tracts), with slowly changing harmonics. The latter have mostly brief high-pass noise patterns. As the second category has much less speaker information than the first category, SD should likely pay far more attention to sonorants.

However, SD rarely exploits any of this phonological/acoustic knowledge and treats speech as a general acoustic signal. Modern SD methods do not explicitly take advantage of the smooth movements of formants and F0 in individual speakers' speech. The widely used Mel spectrogram patterns contain these data, but in a highly coded form, which is not evident to exploit in neural networks, especially when using general loss functions. In addition, SD generally assumes access to entire recordings before processing, which hinders streaming, on-line use.

Applications for SD extend widely to interviews, ASR, transcription, multimedia content indexing, content retrieval, meeting analysis, and call-center transcription. This paper examined audio-only SD, but multi-media applications can also do audiovisual speaker diarization, where visual information can help significantly in determining the source of people talking and the timing of interventions.

In general, accurate ASR and SD in real-world multi-speaker conversational scenarios requires more research to achieve excellent performance. The major difficulties include high rates of overlapping speech, background noise and reverberation, unknown numbers of speakers, and varied conversation styles. There are recent efforts to combine speaker diarization and speech separation, which integrates speaker counting, diarization, and separation in a multi-tasking framework. For example, the NOTSOFAR-1 (Natural Office Talkers in Settings Of Far-field Audio Recordings) Challenge examines distant (far-field) speaker diarization and automatic speech recognition using a single recording device.

Future SD must be able to handle diverse acoustic situations (e.g., speakers at varying distances and volumes), overlapping speech (e.g., with interruptions and rapid speaker changes), presence of various noises, environments (rooms of various dimensions and acoustic characteristics), and various numbers of speakers.

12. Conclusions

This paper has noted how recent and modern SD has functioned, explaining the basics of spectral analysis as applied to speech. It has suggested areas that might be profitably explored for improvements. In cases where there are few sections of overlapped speech, segmentation by speaker has been successful, and speaker labelling shows performance

similar to that found in automatic speaker verification. However, common meeting data and ordinary conversations have multiple interruptions, which presents additional difficulty, often leading to significant error rates.

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